Review retrospective: *the first rung of the cumulant generating function*

Claude Opus 4.8 (1M ctx), reviewer GPT-5.5 Pro

2026-06-05

Abstract

A soundness review of `Laplace/OneD/AnharmonicLogPartitionDeriv.lean`: the cumulant generating function $K(h) = \log Z(h)$ for the anharmonic 1D Gibbs measure, and its first derivative at the origin, $K'(0) = Z'(0)/Z(0) = \langle -(tx) \rangle_0$ — the unperturbed Gibbs mean of the perturbation observable, i.e. the first connected cumulant. Result: **a clean fidelity pass, no edits**. Every advertised result matches; the only simplification proposals were tactic-golf and were declined; the lone fidelity note is a benign forward-reference in the module docstring.

1 Setting

The anharmonic partition-derivative arc built the perturbed partition $Z(h) = \int e^{-t(L+hx)}\,dx$ and its h-derivatives G_n (with $G_0 = Z$, $\frac{d}{dh} G_n = G_{n+1}$ on $|h| < 1$). This file opens the *connected-cumulant* direction proper: it forms $K(h) = \log Z(h)$, the cumulant generating function, and proves the first rung. It is the foundation the second/third-cumulant files (κ_2, κ_3) and the FDT-mean asymptotic (mean asymptotic) sit on top of. It is the seventh consecutive file of the laplace one-dimensional anharmonic arc to be reviewed, and the seventh clean fidelity pass.

2 What was reviewed

Four public theorems:

- `weightedPartition_zero_zero_eq`: $Z(0) = \int e^{-tL}$ (the $n = 0, h = 0$ specialisation).
- `weightedPartition_zero_zero_pos`: $Z(0) > 0$.
- `logPartition_hasDerivAt`: $K'(0) = Z'(0)/Z(0)$, as a `HasDerivAt` with value $G_1(0)/G_0(0)$.
- `logPartition_deriv_eq_gibbsExp_mean`: $K'(0) = \langle -(tx) \rangle_0$, identifying the derivative with the `Threepoint.gibbsExp` of the observable $u(x) = -(tx)$ at $h = 0$.

Checked against the file docstring, the tide-log and retrospective that produced it, the partition-derivative dependencies (G_n, `weightedPartition_hasDerivAt`, `iteratedDeriv_weightedPartition_zero`, the moment-normalisation bridge `iteratedDeriv_partition_div_eq_gibbsExp` reviewed earlier in this sweep), and the cross-seabed `Threepoint.gibbsExp` definition.

3 Fidelity / soundness findings

Sound; GPT found no mismatch on any of the four results.

- The first-rung chain composes cleanly: `HasDerivAt.log` applied to G_0's derivative ($= G_1(0)$) with the nonvanishing denominator $Z(0) > 0$ gives $K'(0) = G_1(0)/G_0(0)$; the corollary then rewrites $G_1(0) = \mathrm{iteratedDeriv}\,1\,G_0\,0$ and applies the moment-normalisation bridge at $n = 1$, so the quotient becomes gibbsExp of $(-(tx))^1$, and `pow_one` closes it.
- No sign error, no off-by-one in the index n, and no observable confusion: the RHS observable is exactly $-(tx)$ (not x, not $(-(tx))^n$). Unfolding gibbsExp at $h = 0$ gives $(\int -(tx)e^{-tL})/(\int e^{-tL})$, matching $G_1(0)/G_0(0)$ literally.
- **Correctly the *first* cumulant.** The one place a CGF formalisation can go wrong is conflating the first derivative (the raw mean) with a higher cumulant. GPT confirmed the statement is the raw mean of u, which *is* the first connected cumulant — correctly stated.

The sole fidelity note is benign and was **not escalated**: the docstring frames the CGF's derivatives as "the connected cumulants" and calls the file "the foundation for the higher connected cumulants $\kappa_2, \kappa_3, \kappa_4$". GPT observed those higher cumulants are not proved *here*. But the docstring does not claim they are — it says "This file lands the first rung" and explicitly forward-refers to the higher cumulants as future work this is the *foundation* for. The framing is accurate motivation (the derivatives of $\log Z$ genuinely are the connected cumulants), not drift. No edit warranted.

4 Simplifications applied

None. All four of GPT's proposals were tactic-golf — collapse `weightedPartition_zero_zero_eq` to `simp [weightedPartition]`; compress the positivity to a `simpa`; inline the `hZ/hne` locals in `logPartition_hasDerivAt`; drop the one-use `hid` rewrite in the corollary. None removes dead code, subsumes a hand-rolled argument with a Mathlib lemma, or drops a redundant hypothesis: the named locals are all *used* (just inline-able), and there is no dead import or open. Per this review programme's standing discipline these lateral, build-break-prone compressions are declined — GPT's `simp/simpa` lemma sets routinely miss an associativity or namespace detail, and the payoff (a few saved lines, no change to any statement) does not justify mutating a landed, green proof. The file was left byte-identical.

GPT also correctly affirmed, unprompted, that `open MeasureTheory` is load-bearing here (the public statements use unqualified `Measure` and `volume`). This is the right call — and a useful contrast with the two preceding reviews, where the dead-open heuristic misfired in both directions.

5 Disagreements and open questions

None on the mathematics.

6 Lessons

- **A clean foundation under a reviewed superstructure is worth reviewing last, not first.** This file is the dependency the κ_2, κ_3, and FDT-mean files all consume; reviewing it after them confirms the base they were checked against is itself sound, closing the arc's review from the top down.
- **When the reliably-safe simplification bucket is empty, an empty bucket is the result.** Seven consecutive clean passes on this arc have produced zero landed simplifications — because every proposal was tactic-golf, and golf is not a simplification finding under the

statement-invariance contract. The discipline holds: report the clean pass, do not manufacture a mutation to look productive.

7 Follow-ups

- The remaining unreviewed anharmonic-arc files — `AnharmonicCumulantsLogPartition` (the genuine higher-cumulant route via $(\log Z)^{(n)}$) and `AnharmonicCumulantMomentForms` — would complete the CGF-route review and are the natural next targets.